THE EXTENDED PRESHEAF IS A SHEAF WHEN THE ORIGINAL PRESHEAF IS A SHEAF FOR THE BASE TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [4583, 4598] and proof(s) [[4600, 4687]]; this spec covers the complete environment instead (statement 4583-4598, proof span(s) [(4600, 4687)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Editorial note. The overlap-agreement verification and the final restriction to the original sheaf are 'similar to the above' same-shape steps; the core descent is written.

Source reference. Original lemma 0EUW: sources/stacks-chow/tags.txt, source lines 18566–18566. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00QN: sources/stacks-chow/tags.txt, source lines 973–973. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 049N: sources/stacks-chow/tags.txt, source lines 5586–5586. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S, $\mathcal{C}$, F satisfy conditions (1), (2), (a), and (b) of Lemma 0EUW and denote $F' : (Sch/S)^{opp} \to Sets$ the unique extension constructed in the lemma. Let $\tau \in \{Zariski, étale, smooth, syntomic, fppf\}$. Assume*

(c) *for any standard τ-covering $\{V_i \to V\}_{i=1,\ldots,n}$ of affines in Sch/S such that $V \to S$ factors through an affine open $U \subset S$ and $V \to U$ is of finite presentation, the sheaf condition hold for F and $\{V_i \to V\}_{i=1,\ldots,n}$*[1].

Then F' satisfies the sheaf condition for all τ-coverings.

Proof. Let X be a scheme over S and let $\{X_i \to X\}_{i\in I}$ be a τ-covering. Let $s_i \in F'(X_i)$ be elements such that s_i and s_j map to the same element of $F'(X_i \times_X X_j)$ for all $i, j \in I$. We have to show that there is a unique element $s \in F'(X)$ restricting to $s_i \in F'(X_i)$ for all $i \in I$.

Special case: X is an affine such that the structure morphism maps into an affine open U of S and the covering $\{X_i \to X\}_{i\in I}$ is a standard τ-covering. In this case we can write

$$X = \lim V_k$$

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[1] This makes sense as V, V_i, and $V_i \times_V V_j$ are isomorphic to objects of $\mathcal{C}$ by (2).

as a cofiltered limit with $V_k \to U$ of finite presentation and V_k affine. See Algebra, Lemma 00QN. By Lemma 049N there exists a k and a standard τ-covering $\{V_{k,i} \to V_k\}_{i\in I}$ whose base change to X is the given covering. For $k' \geq k$ denote $\{V_{k',i} \to V_{k'}\}_{i\in I}$ the base change to $V_{k'}$ of our covering. Then we see that

$$\begin{aligned} F'(X) &= \operatorname{colim}_{k'\geq k} F(V_k) \\ &= \operatorname{colim}_{k'\geq k} \text{Equalizer}(\prod F(V_{k',i}) \rightrightarrows \prod F(V_{k',i} \times_{V_{k'}} V_{k',j}) \\ &= \text{Equalizer}(\operatorname{colim}_{k'\geq k} \prod F(V_{k',i}) \rightrightarrows \operatorname{colim}_{k'\geq k} \prod F(V_{k',i} \times_{V_{k'}} V_{k',j}) \\ &= \text{Equalizer}(\prod F'(X_i) \rightrightarrows \prod F'(X_i \times_X X_j) \end{aligned}$$

The first equality holds by construction of F'. The second holds by assumption (c). The third holds because filtered colimits are exact. The fourth again holds by construction of F'. In this way we find that the sheaf property holds for F' with respect to $\{X_i \to X\}_{i\in I}$.

General case. Choose an affine open covering $X = \bigcup U_k$ such that each U_k maps into an affine open of S. For every k we may choose a standard τ-covering $\{V_{k,j} \to U_k\}_{j=1,\ldots,m_k}$ which refines $\{X_i \times_X U_k \to U_k\}_{i\in I}$. For each $j \in \{1,\ldots,m_k\}$ choose an index $i_{k,j} \in I$ and a morphism $g_{k,j} : V_{k,j} \to X_{i_{k,j}}$ over X. Let $s_{k,j}$ be the element of $F'(V_{k,j})$ we get by restricting $s_{i_{k,j}}$ via $g_{k,j}$. Observe that $s_{k,j}$ and $s_{k',j'}$ restrict to the same element of $F'(V_{k,j} \times_X V_{k',j'})$ for all k and k' and all $j \in \{1,\ldots,m_k\}$ and $j' \in \{1,\ldots,m_{k'}\}$; verification omitted. In particular, by the result of the previous paragraph there is a unique element $s_k \in F'(U_k)$ restricting to $s_{k,j}$ for all j. With this notation we are ready to finish the proof.

Proof of uniqueness of s: this is true because F' satisfies the sheaf property for Zariski coverings and $s|_{U_k}$ must be equal to s_k because both restrict to $s_{k,j}$ for all j. This uniqueness then shows that s_k and $s_{k'}$ must restrict to the same section of F' over (the non-affine scheme) $U_k \cap U_{k'}$ because these sections restrict to the same section over the τ-covering $\{V_{k,j} \times_X V_{k',j'} \to U_k \cap U_{k'}\}$. Thus by the sheaf property for Zariski coverings, there is a unique section s of F' over X whose restriction to U_k is s_k. We omit the verification (similar to the above) that s restricts to s_i over X_i. $\square$